

	Thus the horizontal distance travelled while it is rising is greater than when it is falling.
2(a)	Intermolecular forces between molecules is negligible.
2(b)	For an ideal gas, its internal energy is the sum of the random distribution of microscopic kinetic energies associated with the particles of the system.
2(c)(i)	$\text{Total K.E.} = \frac{3}{2}NkT$ $= \frac{3}{2}(2.5 \times 10^{23})(1.38 \times 10^{-23})(273.15 + 21)$ $= 1520 \text{ J}$
2(c)(ii)	Since the piston is free moving, the pressure stays constant and thus the gas expands as it is heated up. Work done by gas $= p\Delta V$ $= (1.0 \times 10^5)(6.3 \times 10^{-4}) = 63 \text{ J}$ Increase in internal energy $= \frac{3}{2}Nk\Delta T$ $= \frac{3}{2}(2.5 \times 10^{23})(1.38 \times 10^{-23})(15)$ $= 77.6 \text{ J}$ Applying 1st law of thermodynamics, $\Delta U = Q + W$ $77.6 = Q + (-63)$ $Q = 77.6 + 63 = 141 \text{ J}$

$2(eV_i)$	This is so that the waves emitted from both Y and Y' are coherent and the waves